Exam 1 MTH 207 (Fall 2026) — J. Baggett

First Name	Last Name

Before you start. Write your name above and check that you have all 6 pages. All electronics off and out of sight — phones included, even just to tell the time, even in airplane mode. No calculators, notes, or books.

TO EARN POINTS YOU MUST SHOW YOUR WORK. Explain your reasons, justify your steps, and present your logic in an organized way. An answer with no supporting work earns no credit.

You have exactly 55 minutes. When you are told to stop you must stop **immediately** — that is in fairness to everyone. Keep your eyes on your own paper.

(2^{pts} ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) If the average rate of change of f on $[1,3]$ is 0, then f is constant on $[1,3]$.

Solution:

FALSE. Only the endpoints must agree. $f(x) = (x-2)^2$ has equal values at 1 and 3 but is not constant.

(b) $\lim_{x\to 3} \dfrac{x^2-9}{x-3} = \lim_{x\to 3}(x+3)$

Solution:

TRUE. Limits ignore the value at the point, and the two functions agree for $x \neq 3$.

(c) If $\lim_{x\to 0} f(x) = \infty$ and $\lim_{x\to 0} g(x) = \infty$, then $\lim_{x\to 0}[f(x) - g(x)] = 0$.

Solution:

FALSE. $\infty - \infty$ is indeterminate. With $f = \dfrac{2}{x^2}$ and $g = \dfrac{1}{x^2}$ the difference is ∞.

(10^{pts}) **2.** Evaluate the following limits. As a reminder, **justify your steps!** Simplify your answer as much as possible. (If the limit does not exist, state why.)

(a) $\lim_{x\to 3} \dfrac{\frac{1}{x}-\frac{1}{3}}{x-3}$

Solution:

Combine the numerator over a common denominator first.

$$\lim_{x\to 3} \frac{\frac{1}{x}-\frac{1}{3}}{x-3} = \lim_{x\to 3} \frac{\frac{3-x}{3x}}{x-3} = \lim_{x\to 3} \frac{3-x}{3x(x-3)}$$

$$= \lim_{x\to 3} \frac{-(x-3)}{3x(x-3)} = \lim_{x\to 3} \frac{-1}{3x} = \boxed{-\tfrac{1}{9}}$$

The factor $x-3$ cancels because $x \neq 3$ throughout the limit.

(b) Suppose $2x \leq f(x) \leq x^2 - 2x + 4$ for all x. Find $\lim_{x\to 2} f(x)$.

Solution:

Both bounds are polynomials, so their limits come from substitution:

$$\lim_{x\to 2} 2x = 4, \qquad \lim_{x\to 2} \left(x^2 - 2x + 4\right) = 4 - 4 + 4 = 4.$$

The two bounds have the *same* limit, so by the Squeeze Theorem $\lim_{x\to 2} f(x) = \boxed{4}$.

(13pts) **3.** Evaluate the following limits. As a reminder, **justify your steps!** (If the limit does not exist, state why.)

(a) $\lim_{x\to 2} \dfrac{x^2-4}{|x-2|}$

Solution:

Factor first, then split on the sign of $x-2$, since $|x-2| = x-2$ for $x > 2$ and $|x-2| = -(x-2)$ for $x < 2$.

$$\lim_{x\to 2^+} \frac{(x-2)(x+2)}{x-2} = \lim_{x\to 2^+} (x+2) = 4$$

$$\lim_{x\to 2^-} \frac{(x-2)(x+2)}{-(x-2)} = \lim_{x\to 2^-} -(x+2) = -4$$

The one-sided limits disagree, so the limit **does not exist**.

(b) $\lim_{x\to 2^+} \dfrac{x-5}{x^2-4}$, and then say what your answer tells you about the graph of this function.

Solution:

Factor the denominator: $x^2-4 = (x-2)(x+2)$. As $x \to 2^+$ the numerator tends to -3, while $x-2 \to 0^+$ and $x+2 \to 4$, so the denominator tends to 0 through *positive* values. A fixed negative over a shrinking positive gives

$$\lim_{x\to 2^+} \frac{x-5}{x^2-4} = \boxed{-\infty}.$$

What it says about the graph: the line $x = 2$ is a **vertical asymptote**, and just to the right of $x = 2$ the graph plunges downward without bound.

(7pts) **4.** State the Intermediate Value Theorem.

Solution:

The Intermediate Value Theorem. Suppose f is **continuous on the closed interval** $[a, b]$, and let N be any number strictly between $f(a)$ and $f(b)$ (so $f(a) \neq f(b)$). Then there exists a number c in the open interval (a, b) such that $f(c) = N$.

(11 pts) **5.** Find the value of the constant c that makes the function below continuous at every real number. Justify your answer.

$$f(x) = \begin{cases} \dfrac{x^2+x-6}{x-2} & x \neq 2 \\ c & x = 2 \end{cases}$$

Solution:

f is a quotient of polynomials and so is already continuous at every $x \neq 2$. Continuity at $x = 2$ requires $\lim_{x\to 2} f(x) = f(2) = c$.

$$\lim_{x\to 2} \frac{x^2+x-6}{x-2} = \lim_{x\to 2} \frac{(x+3)(x-2)}{x-2} = \lim_{x\to 2}(x+3) = 5$$

So the function is continuous everywhere exactly when $\boxed{c = 5}$.

(9 pts) **6.** Find the horizontal asymptotes of $f(x) = \dfrac{6-4x^2}{3+2x^2}$.

Solution:

Divide numerator and denominator by x^2, the highest power present.

$$\lim_{x\to\infty} \frac{6-4x^2}{3+2x^2} = \lim_{x\to\infty} \frac{\frac{6}{x^2}-4}{\frac{3}{x^2}+2} = \frac{0-4}{0+2} = -2$$

The same computation runs unchanged as $x \to -\infty$, because x^2 is used throughout and $6/x^2, 3/x^2 \to 0$ in both directions. Hence there is one horizontal asymptote, $\boxed{y = -2}$, in both directions.

(11pts) **7.** Evaluate the following limit or state that it does not exist. Show the steps!

$$\lim_{x\to-\infty}\left(\sqrt{4x^2+3x}+2x\right)$$

Solution:
An $\infty-\infty$ form, so multiply by the conjugate, then divide top and bottom by x — the same chain as §2.6. Out at $-\infty$ the variable is **negative**, so $x=-\sqrt{x^2}$, and that one line is the whole difference between $-3/4$ and $+3/4$.

$$\begin{aligned}
\lim_{x\to-\infty}\left(\sqrt{4x^2+3x}+2x\right) &= \lim_{x\to-\infty}\frac{\left(4x^2+3x\right)-4x^2}{\sqrt{4x^2+3x}-2x} = \lim_{x\to-\infty}\frac{3x}{\sqrt{4x^2+3x}-2x} \\
&= \lim_{x\to-\infty}\frac{3x/x}{\dfrac{\sqrt{4x^2+3x}}{x}-\dfrac{2x}{x}} && \text{divide by } x \\
&= \lim_{x\to-\infty}\frac{3}{\dfrac{\sqrt{4x^2+3x}}{-\sqrt{x^2}}-2} && \textbf{here: } x=-\sqrt{x^2} \\
&= \lim_{x\to-\infty}\frac{3}{-\sqrt{\dfrac{4x^2+3x}{x^2}}-2} && \text{one root over one} \\
&= \lim_{x\to-\infty}\frac{3}{-\sqrt{4+\frac{3}{x}}-2} = \frac{3}{-\sqrt{4}-2} = \frac{3}{-4} = \boxed{-\tfrac{3}{4}} && \text{divide inside}
\end{aligned}$$

Using $x=+\sqrt{x^2}$ — true only for $x>0$ — gives $+3/4$. That one line is the whole problem.

(15pts) **8.** Let $f(x)=\sqrt{x+1}$. Use the **definition of derivative** to compute $f'(3)$, and then say what that number represents. Using shortcut rules will earn no points.
Solution:
By the definition, with $f(3)=\sqrt{4}=2$:

$$\begin{aligned}
f'(3) &= \lim_{h\to0}\frac{f(3+h)-f(3)}{h} = \lim_{h\to0}\frac{\sqrt{4+h}-2}{h} \\
&= \lim_{h\to0}\frac{\sqrt{4+h}-2}{h}\cdot\frac{\sqrt{4+h}+2}{\sqrt{4+h}+2} = \lim_{h\to0}\frac{(4+h)-4}{h\left(\sqrt{4+h}+2\right)} \\
&= \lim_{h\to0}\frac{h}{h\left(\sqrt{4+h}+2\right)} = \lim_{h\to0}\frac{1}{\sqrt{4+h}+2} = \frac{1}{2+2} = \boxed{\tfrac{1}{4}}
\end{aligned}$$

What it represents: $f'(3)=\dfrac{1}{4}$ is the **slope of the tangent line** to $y=\sqrt{x+1}$ at the point $(3,2)$ — equivalently, the instantaneous rate of change of f with respect to x at $x=3$.

(7pts) **9.** Give an example of a function f for which $\lim_{x\to 0} f(x)$ does **not** exist but $\lim_{x\to 0} |f(x)|$ **does** exist. Explain why your example works.

Solution:

Take

$$f(x) = \frac{x}{|x|} = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

Why $\lim_{x\to 0} f(x)$ **fails:** the one-sided limits are $\lim_{x\to 0^+} f(x) = 1$ and $\lim_{x\to 0^-} f(x) = -1$. They disagree, so the two-sided limit does not exist.

Why $\lim_{x\to 0} |f(x)|$ **exists:** $|f(x)| = 1$ for *every* $x \neq 0$, so the function is constant near 0 and $\lim_{x\to 0} |f(x)| = 1$.

Taking the absolute value destroys the sign, which was the only thing distinguishing the two sides.

(11pts) **10.** Let $g(x) = x^3 - 4x^2 + 3x + 2e^x$. Find $g'(x)$, then find the equation of the tangent line to $y = g(x)$ at $x = 0$.

Solution:

Differentiate term by term with the Power Rule, and $\frac{d}{dx}e^x = e^x$:

$$g'(x) = \boxed{3x^2 - 8x + 3 + 2e^x}$$

For the tangent line at $x = 0$ we need a point and a slope:

$$g(0) = 0 - 0 + 0 + 2e^0 = 2 \qquad \text{(the point is } (0, 2)\text{)}$$
$$g'(0) = 0 - 0 + 3 + 2e^0 = 5 \qquad \text{(the slope)}$$

Point-slope: $y - 2 = 5(x - 0)$, that is, $\boxed{y = 5x + 2}$.

THIS IS THE LAST PAGE OF THE EXAM.